Midterm Project Specifications

Natural Language Processing (Section: A)

Midterm Project – Summer 25-26

DRAFT

Project Description

The goal of the midterm project is to implement different natural language processing techniques covered so far in this course. This document must be read carefully to complete the project successfully.

The project must be completed in a four-person group. Each group selects two datasets related to one of the following domains: fake news detection, AI content detection, emotion detection, document type detection, and authorship identification. Datasets must not overlap with any other group in the class.

A spreadsheet has been posted in MS Teams to record group and dataset information.

Project Deliverables

Submit a ZIP file (**nlp_mid_project_group_XX.zip**) containing:

- The implemented Python notebook program (**nlp_mid_project_group_XX.ipynb**)
- The project report (**nlp_mid_project_group_XX.pdf**)
- Sample dataset files (first 100 rows of each original dataset)

Replace “XX” in all filenames with the group number — e.g., for group 1, the report filename is **nlp_mid_project_group_01.pdf**. See the instructions below for report details.

During the VIVA session, the implemented program must be brought and may be requested for execution. **The program file must not contain any comments.**

General Instructions

- Ensure group information is complete in the spreadsheet.
- Ensure unique datasets are selected by reviewing the datasets already chosen by other groups in the spreadsheet.
- Use the report template provided in MS Teams to write the project report.
- At the beginning of the report (after the cover page), provide a short description of the selected datasets. For each implemented task, write a paragraph describing how the task was implemented, including the relevant code segment and sample output. Descriptions should include only content sufficient to understand the solution and its associated code segment. After describing the program, describe the result table (below). Provide the full code at the end of the report. No description is required for the pre-task.
- The submission deadline for all deliverables is **July 27, 2026, 11:59 PM**.
- Comments are not allowed in the implemented program.
- The project VIVA schedule will be announced in MS Teams.
- Copying or using AI-generated content from any source is prohibited and will be strictly handled.

Project Requirements

Pre-Task: Provide group and dataset information in the spreadsheet.

Task: Apply the Naïve Bayes, Logistic Regression, and Support Vector Machine algorithms to the selected datasets for classification, and evaluate the learned models using the evaluation metrics covered in class. To complete this task, apply any applicable combination of the following techniques:

- Tokenization
- Case folding
- Synonym substitution
- Stemming
- Lemmatization
- Punctuation removal
- Stop words removal
- Vector Semantics (Bag of Words, TF-IDF)

At the end, complete the following table:

Algorithm	Representation	Dataset 1	Dataset 2
Naive Bayes	BoW		
	TF-IDF		
Logistic Regression	BoW		
	TF-IDF		
Support Vector Machine	BoW		
	TF-IDF		

- Each sub-task must be completed in a separate program cell in the Python notebook file — e.g., one cell to mount Google Drive, another to read the dataset files, and so on. No action is required for the pre-task.
- Once the program is complete, execute the entire code to generate the output. The program must be submitted together with the generated output.
- Scikit-learn supports multiple implementations of these algorithms. This project requires **MultinomialNB** for Naïve Bayes, **LogisticRegression**, and **LinearSVC** for Support Vector Machine. Default configurations may be used for all of them.
- Do not use any library requiring installation other than **nltk**, **spaCy**, **matplotlib**, **numpy**, **pandas**, and **scikit-learn** (or any other library discussed in class).
- An assignment named “Midterm Project” has been created in MS Teams. Submit the ZIP file under that assignment by the deadline.

Submission Deadline: July 27, 2026, 11:59 PM